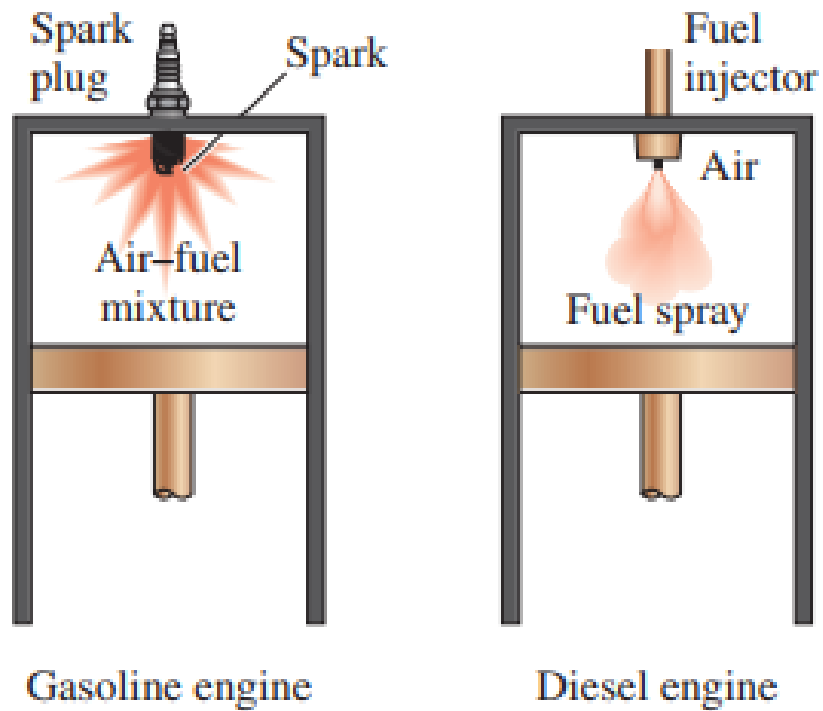


1.3 Diesel Cycle: The ideal Cycle for Compression Ignition Engine or Diesel Engine

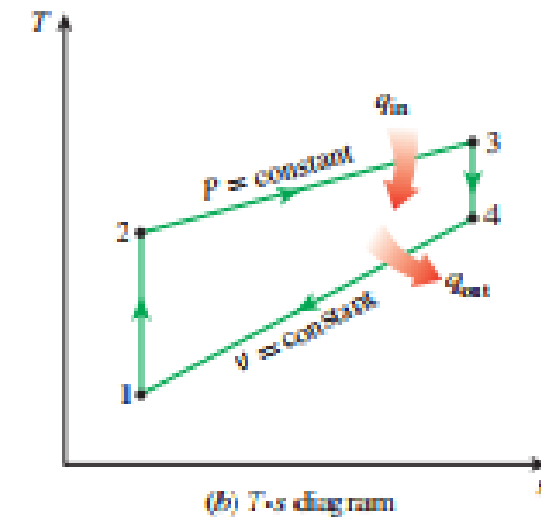
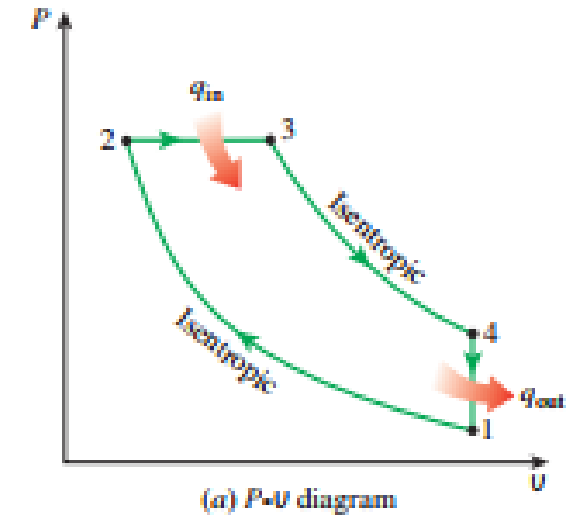
- This cycle was discovered by a German engineer Dr. Rudolph Diesel.
- Diesel cycle is also known as constant pressure heat addition cycle.
- The ideal diesel cycle consists of two reversible adiabatic process, a constant pressure and constant volume process.

Diesel Cycle consists of four processes:

1. **Isentropic Compression (1 – 2)** : Air is compressed, increasing its temperature and pressure.
 2. **Constant Pressure Heat Addition (2 – 3)** : Fuel is injected and burns, adding heat at approximately constant pressure. This drives the piston down.
 3. **Isentropic Expansion (3 – 4)**: The hot, high-pressure gas, performing work.
 4. **Constant Volume Heat Rejection(4 – 1)**: Heat is rejected from the system at a constant volume before the next cycle begins.
- The working fluid is taken in the engine cylinder during the suction stroke process



In the diesel cycle engines, the spark plug is replaced by a fuel injector, and only air is compressed during the compression process



$T -$ and $p - v$ diagrams for the ideal Diesel cycle

Most modern diesel engines operate on a four-stroke cycle:

- 1.Intake Stroke:** The intake valve opens, and the piston moves down, drawing **only clean air** into the cylinder.
- 2.Compression Stroke:** Both valves close. The piston moves up, rapidly compressing the air, causing its temperature to rise significantly (up to 700°C).
- 3.Power (Expansion) Stroke:** As the piston approaches **Top Dead Center (TDC)**, **atomized diesel fuel** is injected. It instantly ignites in the hot, compressed air. The resulting rapid expansion of gases drives the piston down, doing work.
- 4.Exhaust Stroke:** The exhaust valve opens. The piston moves up, pushing the combustion products (exhaust gas) out of the cylinder.

Process 1 – 2: Reversible adiabatic or isentropic compression process

- Reversible adiabatic or isentropic compression of air from state 1 to 2.
- Pressure and temperature of air is increase.

Process 2 – 3: Constant pressure heat Supply

- During this process heat is added to air at constant pressure.
- Due to heat addition volume and temperature increases.
- Volume ration v_3/v_2 is known as cut off ratio (r_c).
- Heat supplied $q_1 = C_p(T_3 - T_2)$

Process 3 – 4: Reversible Adiabatic Expansion

- Reversible adiabatic or isentropic expansion or isentropic expansion of air from state 3 to sate 4.
- Work is developed during this process.

Process 4 – 1: Constant Volume Heat Rejection

- Constant volume heat rejection is carried out during this process.
- Hence pressure and temperature of air decreases to initial value.
- This way the cycle is completed
- Heat Rejected $q_1 = C_v(T_4 - T_1)$

Efficiency of Diesel Engine:

$$\text{Heat supplied } q_1 = C_p(T_3 - T_2)$$

$$\text{Heat Rejected } q_2 = C_v(T_4 - T_1)$$

$$\text{Net work done, } W_{net} = q_1 - q_2$$

$$W_{net} = C_p(T_3 - T_2) - C_v(T_4 - T_1)$$

$$\eta_{Diesel} = \frac{\text{Output}}{\text{Input}} = \frac{\text{Net work}}{\text{Heat Input}} = \frac{C_p(T_3 - T_2) - mC_v(T_4 - T_1)}{C_p(T_3 - T_2)} = 1 - \frac{C_v(T_4 - T_1)}{C_p(T_3 - T_2)}$$

$$C_p / C_v = k$$

$$\eta_{Diesel} = \frac{1}{k} \frac{(T_4 - T_1)}{(T_3 - T_2)}$$

$$\text{Compression Ratio. } r = \frac{v_1}{v_2}$$

$$\text{Cut off Ratio, } r_c = \frac{v_3}{v_2}$$

$$\text{Expansion Ratio, } r_E = \frac{v_4}{v_3} = \frac{\frac{v_4}{v_2}}{\frac{v_3}{v_2}} = \frac{r}{r_c}$$

$$(v_1 = v_4)$$

Now find the relations T_2 , T_3 and T_4

Process 1 – 2

Governing Law is $pv^k = C$

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \right)^{\frac{k-1}{k}} = \left(\frac{v_1}{v_2} \right)^{k-1}$$

$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2} \right)^{k-1} = (r)^{k-1}$$

$$T_2 = T_1 \cdot (r)^{k-1}$$

Process 2 – 3:

Constant pressure Process $p = c$
($p_2 = p_3$)

$$\frac{p_3 v_3}{T_3} = \frac{p_2 v_2}{T_2}; \frac{T_3}{T_2} = \frac{v_3}{v_2}$$

$$T_3 = T_2 r_c$$

$$T_3 = T_1 r_c (r)^{k-1}$$

Process 3 – 4

Governing Law $pv^k = C$

$$\frac{T_4}{T_3} = \left(\frac{p_4}{p_3}\right)^{k-1/k} = \left(\frac{v_3}{v_4}\right)^{k-1}$$

$$\frac{T_4}{T_3} = \left(\frac{v_3}{v_4}\right)^{k-1}$$

$$\frac{T_4}{T_3} = \left(\frac{r_c}{r}\right)^{k-1} \quad \frac{v_4}{v_3} = \frac{r}{r_c}$$

$$T_4 = T_3 \frac{r_c^{k-1}}{r^{k-1}}$$

$$T_4 = T_1 r_c (r)^{k-1} \frac{r_c^{k-1}}{r^{k-1}} = T_1 r_c^k$$

$$\eta_{Diesel} = \frac{1}{k} \frac{(T_4 - T_1)}{(T_3 - T_2)}$$

$$T_2 = T_1 \cdot (r)^{k-1} ; T_3 = T_1 r_c (r)^{k-1} \text{ and } T_4 = T_1 r_c^k$$

Replacing the above into the efficiency equation,

$$\eta_{Diesel} = \frac{1}{k} \frac{(T_1 r_c^k - T_1)}{(T_1 r_c r^{k-1} - T_1 r^{k-1})} = \frac{1}{k} \frac{(r_c^k - 1)}{(r_c r^{k-1} - r^{k-1})} = 1 - \frac{1}{r^{k-1}} \frac{1}{k} \left(\frac{r_c^k - 1}{r_c - 1} \right)$$

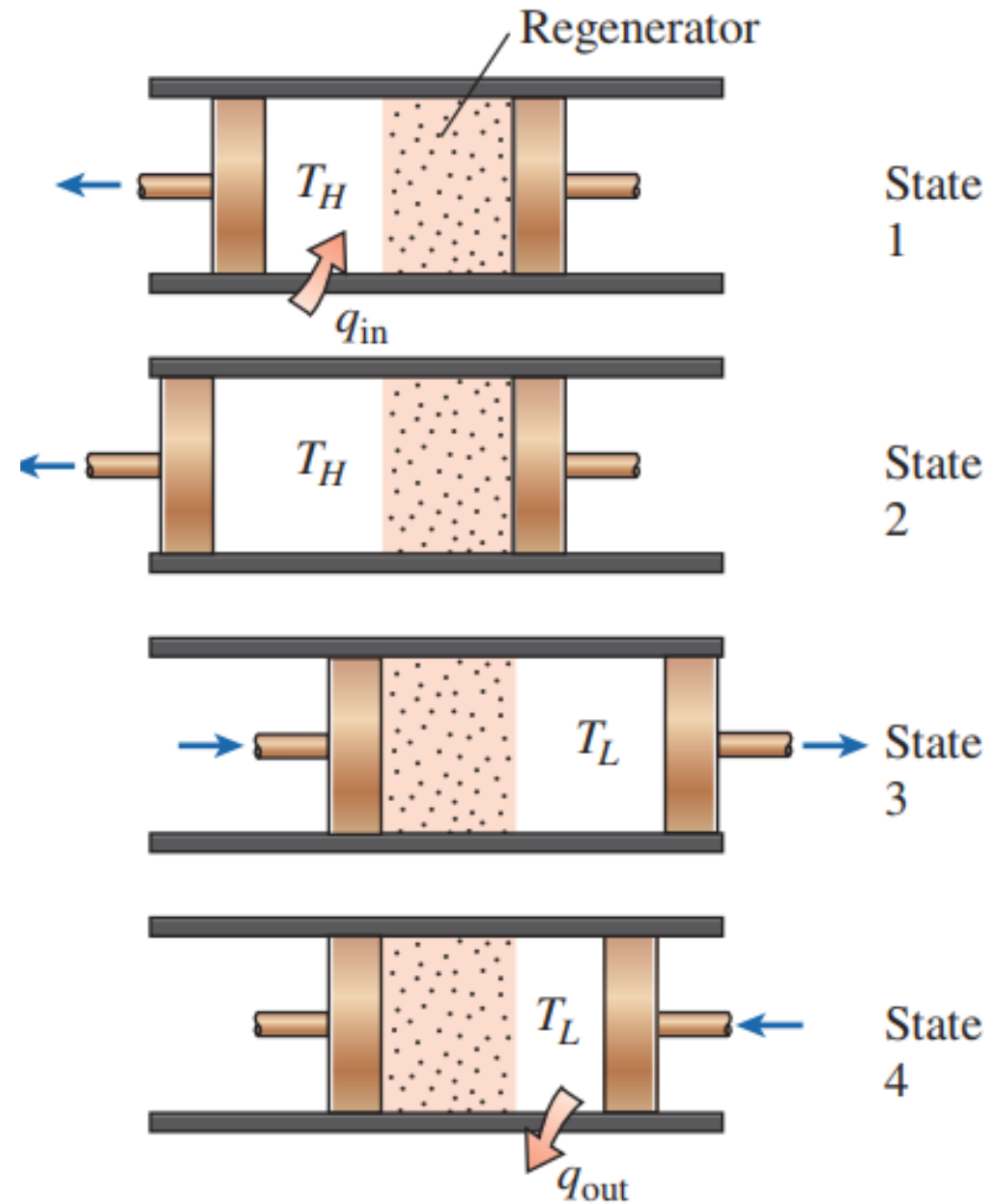
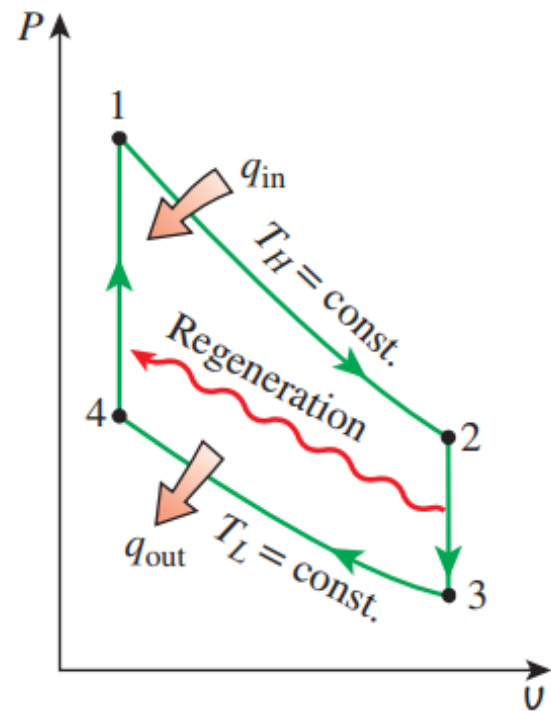
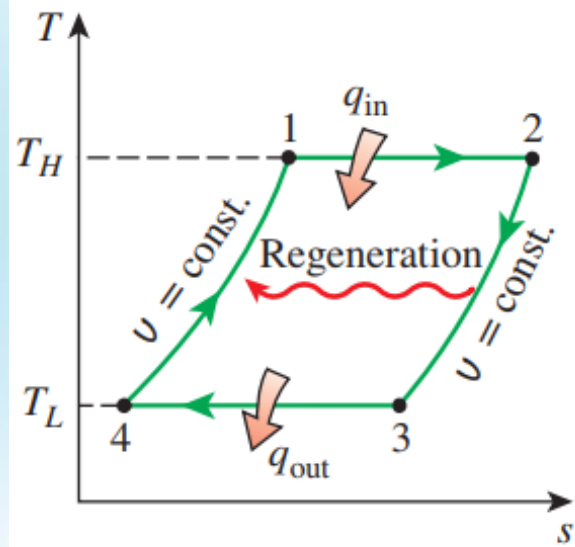
Examples

1. An Air – standard Diesel cycle has compression ratio of 16 and a cutoff ratio of 2. At the beginning of the compression process, air is at 95 kpa and 27°C. Accounting for constant specific heats, determine a) the temperature after heat-addition process, b) the thermal efficiency and c) the mean effective pressure.
2. An Air – standard Diesel cycle has a compression ratio of 16 and a cutoff ratio of 2. At the beginning of the compression process, air is at 95 kpa and 27°C. Accounting for the ***variation of specific heats***, determine a) the temperature after heat-addition process, b) the thermal efficiency and c) the mean effective pressure.

1.4 Stirling and Erickson Engine

1.4.1 Stirling Engines

- Composed of two isothermal and two isochoric (constant volume) processes
- Heat is added and rejected during the isothermal stages.
- A regenerator is used to store heat during the constant volume processes, improving efficiency.
- A regenerator is a device which alternately stores and releases heat energy is a essential component of a Stirling engine.
- The thermal mass (product of mass and specific heat) of a regenerator is is usually high.
- The cycle achieves Carnot efficiency when operating between the same temperature limits.
- Commonly used in low-power applications like cryogenic coolers and solar-powered engines.



The above figures shows the $T - S$ and $p - v$ diagram of the Stirling cycle, which is made up of four totally reversible processes:

1-2 $T = \text{constant}$ expansion (heat addition from the external source)

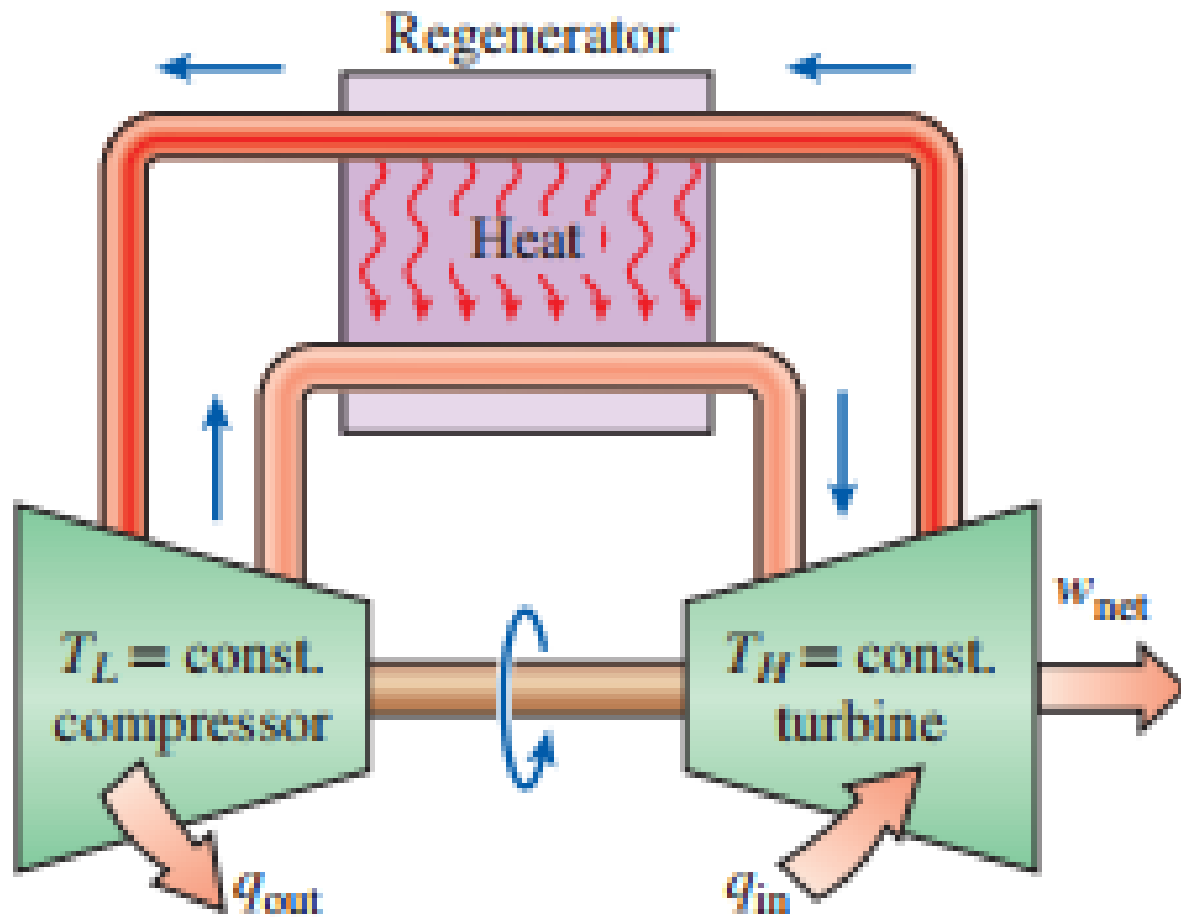
2-3 $v = \text{constant}$ regeneration (internal heat transfer from the working fluid to the regenerator)

3-4 $T = \text{constant}$ compression (heat rejection to the external sink)

4-1 $v = \text{constant}$ regeneration (internal heat transfer from the regenerator)

9.4.2 Ericsson Engines

- The Ericsson cycle is very much like the Stirling cycle, except that the two constant volume processes are replaced by two constant- pressure processes.
- Here the isothermal compression and expansion processes are executed in a compressor and a turbine, respectively, and a counter flow heat exchanger serves as a regenerator.
- Hot and cold fluid streams enter the heat exchanger from opposite ends, and heat transfer takes place between the two streams.



A steady-flow Ericsson Engine

$$\eta_{th,Stirling} = \eta_{th,Ericsson}$$

$$= \eta_{th,Carnot} = 1 - \frac{T_L}{T_H}$$

